

Busy Beaver Calendar

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What we are trying to solve

Goal

Increasing student engagement by transforming the weekly EECS event email into map based experience

Target

OSU EECS Students

Value

Enhanced Discoverability through spatial visualization

Why and the current problem

Lack of visual context

Next Week @ EECS [Summarize](#)

3 attachments (2 MB) [Save all to OneDrive - Oregon State University](#) [Download all](#)

NEXT WEEK – Career Fair Week!
NNL - Hiring STEM Interns & Full-Time
ACM hosts: Feb. 9 @ 12:30 pm – 1:30 pm | KEC 1007 + Zoom | [Register](#)
HPS hosts: Feb. 9 @ 5:00 pm - 6:00 pm | KEC 1007 + Zoom | [Register](#)
HKN hosts: Feb. 10 @ 5:00 pm - 6:00 pm | KEC 1007 + Zoom | [Register](#)
NNL looks for first- and second-year students for internships as well as graduating seniors for full-time jobs. MS/MEng students welcome. Citizenship required.

Fast Enterprises – Interns & Jobs
GDG hosts: Feb. 10 @ 1:00 pm – 2:00 pm | KEC 1005 + Zoom | [Register](#)
ACM-W hosts: Feb. 10 @ 6:00 pm – 7:00 pm | KEC 1005 + Zoom | [Register](#)
Fast Enterprises creates essential software for governments. Citizenship may be required.

Pacific Northwest National Lab Info Sessions - Internships & Jobs
Feb. 10 @ 12:00 pm – 1:00 pm | Johnson 221 | [Register](#)
Feb. 10 @ 6:00 pm – 7:00 pm | Johnson 221 | [Register](#)
Learn how you can make an impact through world-class research that drives national security, clean energy, and scientific innovation.

In-Person Reception for EECS Ecampus Students
Feb. 11 @ 3:00 pm – 4:30 pm | [Register](#)
Ecampus students – this reception is for you! Stop by after the OSU STEM Career Fair and meet industry and your faculty.

STEM Industries Career and Internship Fair
Feb. 11 @ 11:00 am - 3:00 pm | CH2M Hill Alumni Center | [Register](#) | [Internships & Jobs](#)
Network with employers and grad schools seeking Oregon State students and recent grads in the fields of Computer Science, Engineering and Math.

Civil, Construction & Architecture Engineering Career Fair – Wants EE
Feb. 11 @ 11:00 am - 3:00 pm | Reser | [Register](#) | [Internships & Jobs](#)
Several employers will hire EE. Check Handshake.

Willamette Valley Society of Women Engineers Reception
Feb. 11 @ 4:30 pm – 7:00 pm | LaSells Stewart Center | [Register](#)
A night of networking, food, scholarship presentations, and an engaging keynote speech from OSU professor Dr. Nordica MacCarty.

Pearl Harbor Navy Info Session – Hiring!
Feb. 11 @ 5:00 pm – 6:00 pm | KEC 1005 | [Register](#)
Learn more about employment opportunities at Pearl Harbor Naval Shipyard and Intermediate Maintenance Facility in Hawaii.

Events

[All Corvallis](#) [Honors](#) [Engineering](#)

[All Campus Events](#)

Career Development

Engineering Connections Virtual Fair

October 7th, 4:00pm - 6:00pm

Save the date for the Engineering Connections Fair

The Engineering Connections Fair is a targete...

Career Development

2025 Fall Civil, Construction & Architectural Engineering Fair

October 14th, 11:00am - 3:00pm

We look forward to seeing you at the Fall Civil, Construction & Architectural Engineering Fair on 10/14 from 11 a.m.- 3p.m.

Career Development

2025 Engineering Connections Fair

October 14th, 4:00pm - 8:00pm

[Events](#) [Resources](#) [Settings](#)

Our Solution

Busy Beaver Calendar (Mock)

Static HTML + CSS only (local mock page)

Calendar (example)

If needed, JavaScript can be added later. For now, this is a simple static table.

Sun	Mon	Tue	Wed	Thu	Fri	Sat
26	27	28	29	30	31	1
2 ●	3 ●	4 ●	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	1

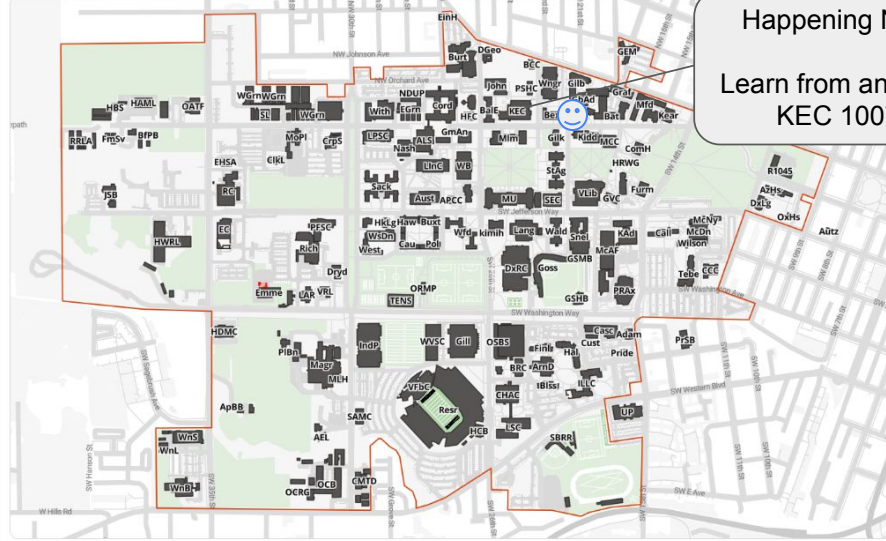
Events (example)

- Feb 2 — Learn from an Alum (KEC 1007)
- Feb 3 — Getting Hired in the Semiconductor Industry (MU Ballroom)
- Feb 4 — OSU Winter Virtual Career & Internship Fair (Virtual)

Local-only mock page.

Event Map (image)

Base map: [OSU Campus Map](#)



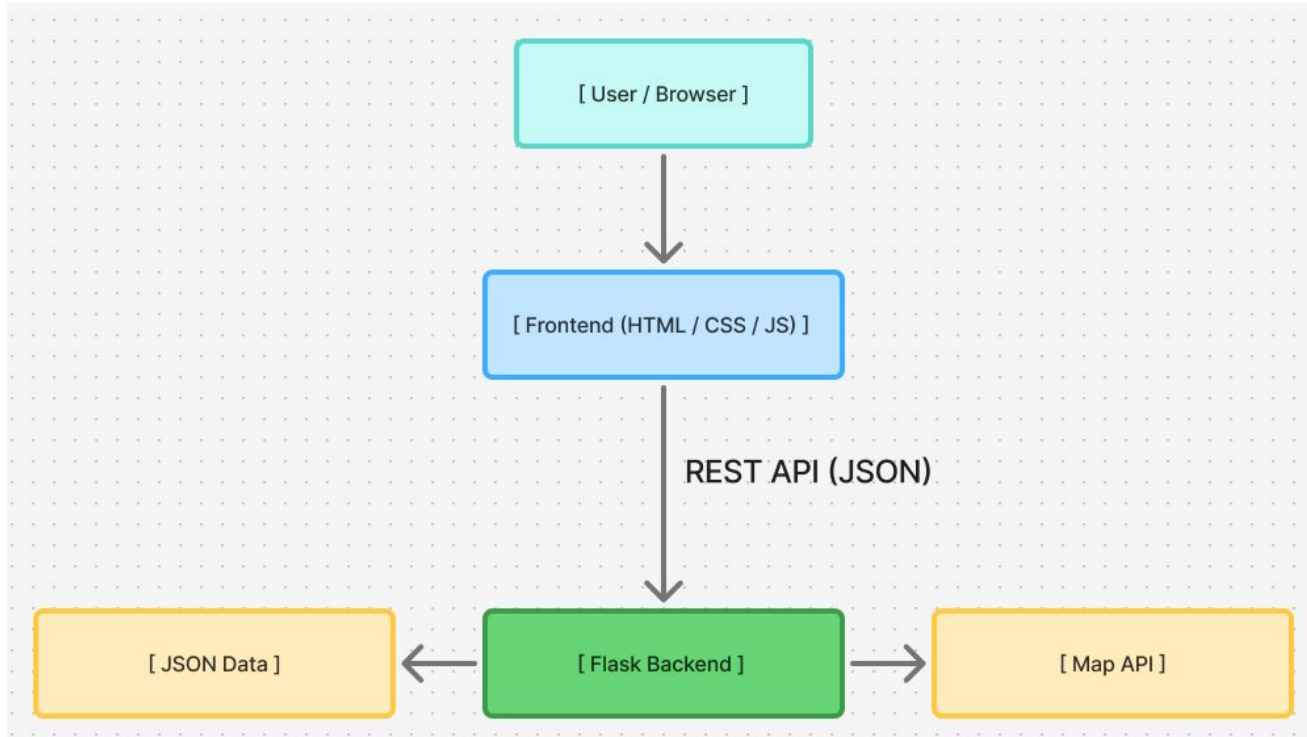
Happening Now

Learn from an Alum
KEC 1007

System Architecture Overview

- Web-based client-server architecture
- Frontend handles UI and user interactions
- Flask backend handles logic and data access
- Event data stored in JSON
- External map API used for visualization

High-Level Architecture Diagram

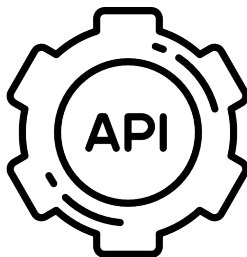


Major Software Components



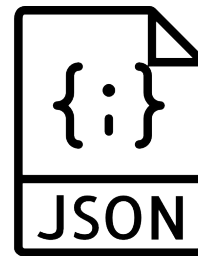
Frontend

- Map
- Calendar
- Filters



Backend

- Event API
- Validation



Data Layer

- Json event storage

Core Features

- Interactive map of OSU campus
 - Visual representation of where EECS events are on a campus map
 - Helps with understanding quickly where an event is
- Event detail
 - Clicking a pin will show title, time, building, and room
- Calendar view
 - Events displayed in a calendar format
- Time-based filtering
 - Ability to filter events by today, this week, or a custom date range
 - Will update both the calendar and the map with correct information
- Fallback behavior
 - If the map API is down for any reason, the events are still viewable in the calendar

Architecture Reasoning

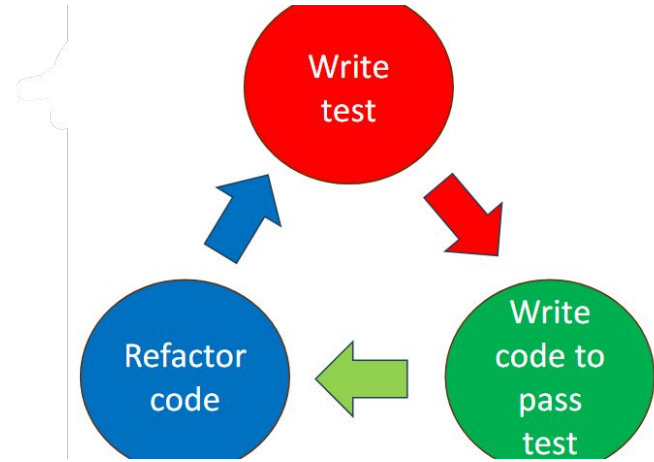
- Client-Server web architecture
- Flask backend for speed
 - Lightweight framework for fast development
- JSON storage for simplicity
 - Choosing this over a database to reduce setup complexity
- RESTful API communication
 - Frontend and backend will communicate via JSON over HTTPS
- External map API integration
 - Offloads map rendering to a third-party service (Google Maps or OpenStreetMap)
- Read-heavy optimization
 - Architecture and UI optimized for lots of readers and few writers

Event Data flow

1. Admin adds an event
2. Backend validates and stores data
3. Frontend fetches events
4. Map renders pins
5. User views details

Methods to test the solution

- We will use a white box testing method. Since we know the content of the code.
- To ensure we reach all of the code We will use Statement and branch coverage.
- And Test Driven Development
- Test inputs from all different subdomains
- Test boundaries of subdomains
- Mutation Analysis



Likely areas to test

- Max number of events that can be added to a day
- Create account that already exists
- Multiple events going on at the same place

How will we know are solution is right

- The user is able to login into their account(s) and no one else's
- Their events are displayed on the calendar and the locations appear on the map
- They can search events and filter them to displayed or not